

Risk Premia in Inflation-Linked Bonds

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RQ1. *BEI residual premia*

US vs UK breakeven minus projected and spot inflation. Similar? If different, why?

RQ2. *Inflation surprises*

TIPS yields react to surprises via protection demand and segmentation. Same for linkers?

RQ3. *Return predictability*

Slope vs CP vs Cieslak–Povala; are forwards cointegrated, and does cointegration predict?

Data & specification

- Curves: **BoE** (UK), **GSW** (US, official Fed files), monthly 2004–2026
- Projected = inflation swap (**ISR**); spot = YoY
- **Linker–swap basis**
 $b(T) \equiv \text{BEI} - \text{ISR} = -\lambda(T)$;
 $b < 0$: cash linker *cheap* to the synthetic
- Surprise = YoY – *full* 120m MA; SA CPI (US), RPI (UK)
- **Actual CPI release dates**
(ECO_RELEASE_DT): US 270, UK 272 months
- t -NW(12) on b ; t -HH(12) on 12m excess returns

RQ1: The Answer Is *Maturity-Dependent* — Shape, Not Level

Linker–swap basis $b = \text{BEI} - \text{ISR}$, common sample 2004–2026 ($n \approx 265$). Last two rows: the **basis differential** $b_{US} - b_{UK}$ and the share of months in which it is positive.

Maturity	3y	5y	7y	10y	15y	20y	25y
b_{US} [bp]	−24	−25	−23	−24	−23	−23	n/a
b_{UK} [bp]	−28	−30	−27	−18	−6	+1	+3
$b_{US} - b_{UK}$ [bp]	+3	+5	+3	−6	−18	−24	—
months $b_{US} > b_{UK}$	53%	67%	63%	30%	8%	2%	—

25y: **UK only** — GSW publish the TIPS curve for integers 2–20, so the comparison stops at 20y. BoE and BPSWIT both cover 25y.

- **Indistinguishable to 7y**: both ≈ -25 bp, sign of the differential a coin toss (53–67% of months) — linkers cheap to the synthetic in both (**FLL 2014**).
- **Sharply different beyond 10y**: the US basis is *flat* across the curve, the UK basis *rises to zero*. At 20y the UK linker is fairly priced to the synthetic; the US is still -23 bp.
- By 20y the sign is **systematic, not episodic**: the US basis is below the UK in 98% of months. The share is distribution-free — it does not depend on how one models the persistence of the differential.
- **Where the UK can be read further, it keeps going**: the basis does not flatten at 20y — it reaches $+3$ bp at 25y and is positive in two months out of three.

RQ1: The Same Comparison, Subtracting *Spot* Inflation

Same object as before, with realised YoY inflation in place of the swap rate. Levels for each market, then the differential.

Maturity	3y	5y	7y	10y	15y	20y
$(\text{BEI} - \text{spot})_{US}$ [bp]	-81	-66	-57	-47	-40	-38
$(\text{BEI} - \text{spot})_{UK}$ [bp]	-88	-82	-76	-63	-45	-36
difference US-UK	+8	+17	+19	+16	+5	-1
months with US>UK	44%	46%	46%	45%	45%	43%

- **Both markets look far cheaper** on this measure (-38 to -88 bp against ≈ -25 on the basis): over the sample realised inflation ran *above* what the swap market projected, and that gap enters the residual.
- **No maturity discriminates**: the sign is a coin toss at every point of the curve (43–46%, against 2–67% on the basis) — which is not the same as the markets agreeing. The identity $\text{BEI} - \text{spot} = (\text{BEI} - \text{ISR}) + (\text{ISR} - \text{spot})$ shows why: at 20y the basis differential is -24 bp and the expectation-gap differential $+22$ bp. **They cancel — two differences, not none.**

RQ1 Why (i): The Aggregate Hides Two Opposite Regimes

Basis differential $b_{US} - b_{UK}$ by sub-period [bp]. The short end **changes sign** in 2009; the long end does not.

Period	3y	5y	7y	10y	15y	20y	25y
<i>Differential $b_{US} - b_{UK}$</i>							
pre-2009	-29	-32	-27	-22	-19	-23	—
2009–2019	+11	+15	+10	-2	-18	-27	—
2020–today	+11	+15	+12	-2	-17	-18	—
<i>UK basis b_{UK} (readable to 25y)</i>							
pre-2009	-12	-9	-8	-5	-2	-2	-3
2009–2019	-36	-38	-34	-24	-10	-0	+5
2020–today	-25	-30	-28	-16	-0	+4	+6

- The insignificant 3–7y differential on the full sample is the **average of two opposite regimes**, not evidence of no difference.
- **Pre-2009** the US basis is more negative at every maturity (TIPS the more dislocated market in the GFC); **post-2009** the ranking inverts at the short end only, and stays inverted.
- **The long-end shape is itself post-2009**: before the crisis the UK basis is *flat* across the whole curve (–12 to –3 bp, 3y to 25y). The upward slope that distinguishes the UK from the US does not pre-exist — **it appears after 2009 and strengthens**, exactly where the LDI clientele operates.
- Event study confirms a UK-specific shock: in the **2022 LDI crisis** the differential moves +56 bp (5y) and +36 bp (10y) versus its own median.

RQ1 Why (ii): Channels, Estimated Separately per Market

Levels regressions of each market's basis on each channel; Newey–West t with *automatic* bandwidth (NW 1994) in brackets; I(1) channels confirmed in first differences. A channel significant in *both* markets moves the two together and cannot explain the differential — hence the third column.

Channel	US	5y UK	difference	US	10y UK	difference
On/off-run	−0.8 [−0.4]	+4.4 [+1.4]	−5.3 [−1.7]	+3.8 [+1.5]	+7.4*** [+3.7]	−3.6 [−1.6]
MOVE	−1.8 [−1.3]	−8.6** [−2.1]	+6.8 [+1.5]	+0.8 [+0.6]	−4.5 [−1.5]	+5.3** [+2.4]
VIX	−2.4 [−1.4]	−2.8 [−0.8]	+0.4 [+0.2]	+0.2 [+0.1]	−1.6 [−0.7]	+1.8 [+0.9]
Noise (HPW)	−24.6*** [−6.8]	+9.7* [+1.8]	−34.3*** [−5.5]	−14.5*** [−3.6]	−1.1 [−0.3]	−13.4*** [−3.3]
HKM capital	+485*** [+4.7]	+453** [+2.0]	+32 [+0.2]	+255* [+1.9]	+297* [+1.8]	−42 [−0.3]

- **Noise is the one channel that separates the two markets** — significant on the difference at both maturities: Treasury-market dislocation cheapens *TIPS* specifically, not gilts.
- **Intermediary capital (HKM) moves both together**: sizeable in each market, zero on the difference — a common global factor, not the explanation.

Expected Inflation: One Measure per Market, Chosen on an External Criterion

RR's US candidates (Cleveland, AR(1), doubly mean-reverting Vasicek) need a UK counterpart. The criterion is **external**: which measure forecast *realised* average inflation over the next T years? RMSE [pp p.a.], common sample, Diebold–Mariano with HAC variance; RW = current YoY.

Horizon	Cleveland	US (1981–2024)			Affine	UK (1992–2025)		
		AR(1)	Vasicek2	RW		AR(1)	Vasicek2	RW
1y	1.24	1.08	1.14	1.14	1.55	1.52	1.56	1.56
3y	1.29	1.29	1.50	1.50	1.53	2.13	2.35	2.35
5y	1.05	1.12	1.40	1.40	1.19	1.61	1.82	1.82
10y	0.80	0.93	1.39	1.40	0.65	1.34	1.58	1.59

- **US: Cleveland** (Haubrich–Pennacchi–Ritchken 2012, the Fed's official series): lowest RMSE from 3y; the gap to the AR(1) is not significant ($DM \leq 1$), DMRV and RW are significantly worse from 5y (DM 2–3). Chosen: Cleveland; AR(1) as the symmetric robustness.
- **UK: the affine model of this paper** (next slide): best at every horizon from 2y, at 10y *half* the AR(1)'s RMSE (DM > 2 from 3y). No UK “Cleveland” exists as a live series; the race shows the affine model is not a weaker substitute for it.
- **One measure per market**, not the best at each maturity: mixing models across maturities would build jumps into the term structure of premia by construction.
- Caveats: full-sample parameters (all models); surveys are inputs of the affine model; few independent windows beyond 5y ($n_{\text{eff}} < 6$).

UK: A No-Arbitrage Decomposition of Breakevens (Kaminska et al., BoE)

The model Liu–Vangelista–Kaminska–Relleen (BoE SWP 551 / JBF 2018), the UK reference: gilt *and* swap breakevens, 3–25y, monthly 1992–2026.

- States: 3 CPI factors + expected **RPI–CPI wedge** + **gilt liquidity** (loads on gilts only, identified by the gilt–swap gap: the LDI channel).
- Kalman filter with surveys as extra measurements: HM Treasury and BoE professional forecasters 1997–2026 (1–4y), Citi/YouGov households 5–10y with an *estimated* bias (+1.2pp) as the long anchor (Consensus is not available).
- **RPI reform 2030** (RPI → CPIH) in the loadings, priced from Sep-2019: without it, post-2020 long breakevens are misread as expected inflation.

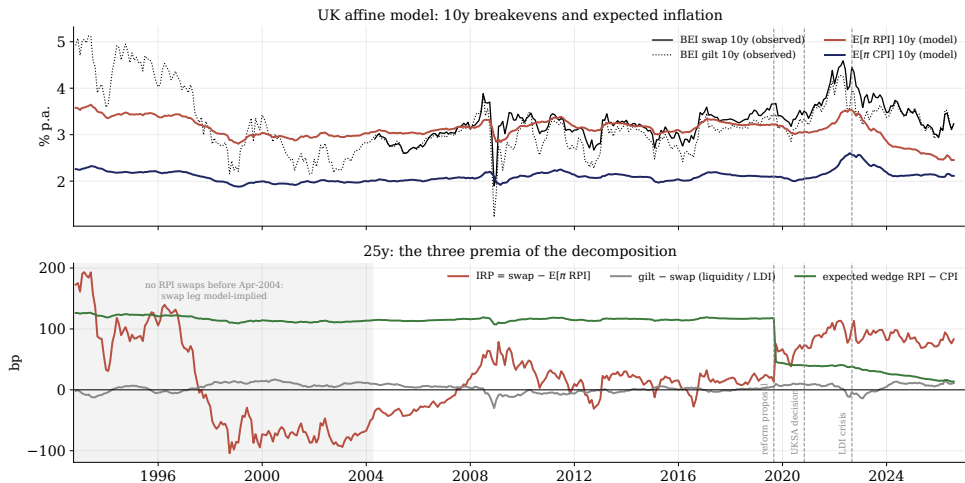
Out-of-sample check BoE Market Participants Survey, professional CPI 5–10y ahead (27 rounds, 2023–26, not used in estimation): survey 2.21%, model 2.15%, s.d. of the gap 0.03pp.

Decomposition of the swap breakeven, sample averages [% / bp]

	5y	10y	20y	25y
<i>2009–2019</i>				
$E[\pi^{CPI}]$	2.08	2.10	2.13	2.14
expected wedge	+98	+108	+113	+114
$IRP = \text{swap} - E[\pi^{RPI}]$	–2	+6	+18	+19
gilt – swap	–32	–18	–5	–1
<i>2020–today</i>				
$E[\pi^{CPI}]$	2.26	2.20	2.18	2.18
expected wedge	+107	+76	+38	+30
$IRP = \text{swap} - E[\pi^{RPI}]$	+48	+64	+83	+84
gilt – swap	–25	–12	+1	+6

- Long-run CPI expectations **anchored at 2.1–2.2%** throughout; 2021–23 is a 3–5y event.
- After 2020 the 25y wedge falls 114 → 30bp (reform) and an **IRP of ≈ 85bp** appears at the long end; without the reform: wedge 117, IRP –5 — same prices, wrong split.
- gilt – swap: negative to 10y, **zero or positive from 20y** — the RQ1 shape as a state.

UK Affine Model: Expectations and Premia over Time



Top: observed 10y breakevens against the model's expected RPI and CPI inflation. Bottom, 25y: the three premia; before April 2004 there are no RPI swaps, so the swap leg is model-implied (shaded). Dashed lines: reform proposal (Sep-2019), UKSA decision (Nov-2020), LDI crisis (Sep-2022).

RQ1 Revisited: Subtracting the *Best Expected Inflation* in Each Market

$BEI - E[\pi] = rp_{\pi} - \lambda$ (total premium in the breakeven) and $ISR - E[\pi] = rp_{\pi}$ (premium in the swap), sample averages 2004–2026 [bp], release-date sampling. US: Cleveland; UK: affine model.

	3y	5y	7y	10y	15y	20y	25y
$(BEI - E[\pi])_{US}$	−4	+6	+13	+17	+16	+11	—
$(BEI - E[\pi])_{UK}$	−12	−16	−11	+4	+25	+36	+37
US−UK $[t-NW]$	+8 [0.9]	+23 [2.6]	+24 [2.5]	+13 [1.3]	−9 [−0.8]	−25 [−2.2]	—
$(ISR - E[\pi])_{US}$	+20	+31	+37	+41	+39	+33	—
$(ISR - E[\pi])_{UK}$	+11	+10	+13	+19	+29	+33	+33

- **Both swap markets carry a positive inflation risk premium** (+10 to +40bp), rising with maturity in the UK and hump-shaped in the US: RR's objection to the swap as “projected inflation” is quantified, and it is of the same order in the two markets.
- **The RQ1 shape survives the change of measure**: the UK total premium rises from −16 to +37bp along the curve while the US stays within ± 17 bp. The differential is positive at 5–7y (+23bp, t 2.5: the US carries the larger premium at the front) and negative at 20y (−25bp, t −2.2) — the same sign and size as on the basis (−25bp), where it is measured without any $E[\pi]$.
- The naive measures (AR(1), Vasicek2) put both markets at −30 to −105bp: they overstate expected inflation by construction (backward-looking), and the US−UK difference on them is never significant. The premium levels depend on the measure; the **shape** does not.

RQ2: Same for Liquidity, *Opposite* for Surprises

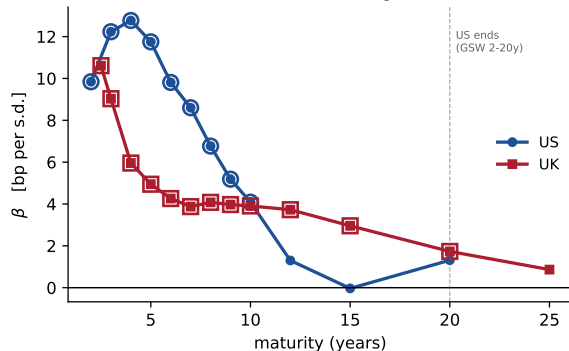
Maffei's specification exactly: *separate* regressions (eq. 7), release-date sampling, surprise = YoY – full 120m MA, liquidity = PC1{VIX, MOVE, noise, on/off}. $\lambda = -b$; bp per s.d. (liq.) or per unit (surp.); t -NW(12) below each coefficient. **25y is UK only**: GSW publishes the TIPS curve for integers 2–20.

	2y / 2.5y	5y	10y	12y	15y	20y	25y
US $\lambda \sim \text{PCA}$	+9.8*** [+3.8]	+11.8*** [+4.9]	+4.1** [+2.5]	+1.3 [+0.9]	−0.0 [−0.0]	+1.3 [+0.8]	— n/a
UK $\lambda \sim \text{PCA}$	+10.6*** [+4.6]	+4.9*** [+2.9]	+3.9** [+2.1]	+3.7** [+2.1]	+3.0** [+2.1]	+1.7** [+2.3]	+0.9 [+0.9]
US $\lambda \sim \text{InfS}$	−1.7 [−1.2]	−1.5 [−0.8]	−1.2 [−1.0]	−1.2 [−1.2]	−1.5* [−1.7]	−2.1** [−2.4]	— n/a
UK $\lambda \sim \text{InfS}$	+6.2*** [+2.6]	+2.9* [+1.9]	+1.0 [+0.8]	+0.3 [+0.3]	−0.2 [−0.2]	+0.3 [+0.9]	+1.0*** [+2.8]

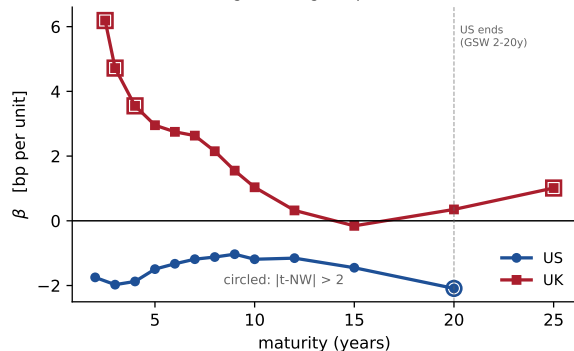
- **Liquidity: the same.** Both load on the short end and fade into the long end (US 11.6→0.9; UK 8.5→1.9 bp, short vs long block). **Maffei's segmentation replicates, and the UK matches it.**
- **Surprises: not the same.** US λ falls, strongest at 20y; UK λ *rises* at the short end — **opposite sign, opposite end.**
- **Beyond the US range the UK changes regime:** at 25y liquidity stops mattering (t 0.9) while the surprise channel *returns* significant with a positive sign (+1.0, t 2.8) — a U-shaped profile, not a decay.
- Validation (Maffei 7.1.2): $\text{ISR} \sim \text{PCA} \approx 0$ in both ($|t| \leq 1.5$) — not a measurement artefact.

RQ2: The Two Channels Across the Whole Curve

$\lambda \sim$ liquidity (PC1): SAME pattern
both fade into the long end



$\lambda \sim$ inflation surprise: OPPOSITE
US negative long, UK positive short



All twelve tenors (the table shows six). **Left:** the liquidity channel is common — both curves decay. **Right:** the surprise channel is not — the UK reacts where its basis matches the US, and is inert at the long end where the two bases diverge.

RQ2: How Much Survives Correct Standard Errors?

The surprise contains a **12-month overlapping window** (YoY), so residual autocorrelation is mechanically induced to lag 11 before any economic persistence. OLS standard errors are badly downward-biased; NW(12) is the minimum defensible truncation.

US $\lambda \sim \text{InfS}$	5y	10y	12y	15y	20y
β [bp/unit]	-1.49	-1.19	-1.15	-1.45	-2.09
t -OLS (<i>as published</i>)	-2.1	-2.9	-3.2	-4.0	-5.5
t -NW(12) (<i>corrected</i>)	-0.8	-1.0	-1.2	-1.7	-2.4

- The t is **stable across lag choice** (6, 12, 18 give -3.1, -2.9, -2.9 at 20y on Maffei's sample): the gap is **OLS vs HAC**, not the truncation.
- **The long-end result survives**: 20y remains significant at 5%. What does not survive is its *strength* — roughly half the published t .
- This strengthens the finding rather than weakening it: the effect is real, and now stated with inference a referee will accept.

RQ2 with Expected Inflation: Do Surprises Move the *Premium*, Not Only the Basis?

Maffei's surprise test is on $\lambda = \text{ISR} - \text{BEI}$, where the inflation risk premium cancels; his $E[\pi]$ enters only the liquidity check on $\hat{\gamma} = \text{BEI} - E[\pi]$ (eq. 7). With the gold-standard measures we can also ask whether surprises move the premium itself: $\text{ISR} - E[\pi]$ on InfS, US Cleveland, UK affine model [bp per unit of surprise, t -NW(12)].

		2–2.5y	3y	5y	7y	10y	15y	20y	25y
US	λ (basis)	–1.7 [–1.2]	–2.0 [–1.1]	–1.5 [–0.8]	–1.2 [–0.8]	–1.2 [–1.0]	–1.5 [–1.7]	–2.1 [–2.4]	—
	$\text{ISR} - E[\pi]$	+15.1 [3.6]	+12.6 [3.5]	+8.2 [2.9]	+5.0 [1.9]	+2.0 [0.8]	+0.3 [0.1]	–0.3 [–0.1]	—
UK	λ (basis)	+6.2 [2.6]	+4.7 [2.2]	+3.0 [1.9]	+2.6 [1.7]	+1.0 [0.8]	–0.2 [–0.2]	+0.4 [0.9]	+1.0 [2.8]
	$\text{ISR} - E[\pi]$	+2.9 [1.2]	+2.4 [1.1]	+2.0 [1.2]	+2.0 [1.3]	+3.1 [1.8]	+3.4 [1.8]	+3.6 [2.0]	+3.6 [2.1]

- **US: surprises are priced in the short-end premium** (+15bp per unit at 2y, t 3.6, gone by 10y) and in the long-end basis (20y, Maffei's TIPS-demand reading). Two objects, two ends of the curve.
- **UK: the premium barely moves** (+2 to +4bp, significant only at 20–25y); the surprise channel is the short-end basis. Consistent with a structure-driven long end.
- The $\hat{\gamma}$ liquidity check (Maffei Table VIII) holds with the new measures: $\hat{\gamma} \sim \text{PC1}$ has the sign and decay of $-\lambda \sim \text{PC1}$ in both markets (UK affine: –22bp/s.d. at 2.5y, t –3.2, to zero by 15y).
- With an AR(1) $E[\pi]$ these regressions are mechanically contaminated (surprise and expectation share the realised YoY: $\beta \approx -50$ to -70 , $t \approx -20$) — the reason the basis, not $\hat{\gamma}$, is the headline object.

RQ3: Two Regimes — Inflation (US) vs Cointegration (UK)

12m real excess returns (average maturity); R^2 with t -HH(12) in brackets; Johansen on 1y real forwards.

	SLOPE	CP	Cieslak–Povala	COINT	Realised infl.	Cointegration
US	0.00 [0.1]	0.22** [2.5]	0.39*** [4.0]	0.11*** [2.9]	0.25*** [−4.2]	$r=4$, ADF −7.9
UK	0.03 [1.0]	0.19*** [3.1]	0.10 [1.6]	0.17*** [4.4]	0.11** [−2.2]	$r=2$, ADF −11.7

Does CiP beat slope and CP? In the US, yes (0.39 vs 0.22 and 0.00) — the inflation-aware factor dominates, consistent with the TIPS result. In the UK, no: CiP is the *weakest* of the three informative factors.

Is the process cointegrated, and does it predict? Yes in both: Johansen finds $r=4$ (US) and $r=2$ (UK), the first ECM residual is strongly stationary, and the cointegration predictor works — it is the **best single factor in the UK** (t 4.4, the highest in either market). Slope alone is inert everywhere.

Realised inflation (Rebonato 2024, JFI) The latest published YoY rate predicts with a *negative* sign in both markets and survives next to every other factor (US: t −4.6 with CP, −2.8 with CiP, CiP+infl. R^2 0.47; UK: t −2.0 to −2.4, with COINT and CP still dominant). The US is the inflation-driven market on this criterion too.

RQ3: Is the Result an Artefact of the Forward Grid?

What is being varied. CP and the cointegration predictor are both built from a vector of **1y forward rates** — CP as a linear combination, COINT from the Johansen system on the same vector. *Which* forwards enter is a free choice, so the test is whether the result depends on it. CiP is built from the inflation trend and cycle, does not use forwards, and is **unchanged by construction** — shown as a reference line. r = Johansen rank (it moves with the grid because the dimension of the system moves).

Forward grid	UK CP	UK COINT	UK r	US CP	US COINT	US r
baseline	0.19***[3.1]	0.17***[4.4]	2	0.22**[2.5]	0.11***[2.9]	4
(4,5,6,7,8)	0.24***[3.2]	0.22***[4.7]	3	0.22**[2.5]	0.11***[3.3]	3
(5,7,10)	0.24***[3.3]	0.22***[5.0]	1	0.14 [1.5]	0.06 [1.3]	1
(4,6,8,10,12)	0.26***[3.0]	0.24***[4.0]	1	0.23***[2.7]	0.11***[4.0]	2
(5,10,15)	0.24***[2.7]	0.23***[3.5]	1	0.14 [1.4]	0.04 [1.3]	1
<i>CiP (reference)</i>	<i>0.10 [1.6]</i>			<i>0.39***[4.0]</i>		

- **UK: no artefact.** COINT holds on every grid (R^2 0.17–0.24, t 3.5–5.0), always paired with CP — consistent with CP *being* a cointegrating combination.
- **US: the forward-based factors are grid-sensitive** — both weaken on sparse grids (5,7,10) and (5,10,15), where Johansen finds a single cointegrating vector. CiP is unaffected and dominates throughout: the US conclusion does not rest on the grid choice.

What the data say

- **RQ1** Same *level* (≈ -25 bp to 7y), **different shape**: the US basis is flat, the UK rises to zero by 20y ($t-11$). Noise — Treasury dislocation — is the only channel that separates them; intermediary capital moves both. The shape survives the best $E[\pi]$ in each market (Cleveland; the UK affine model).
- **RQ2** **Same** for liquidity (short-end loading, fading long — Maffei replicates and the UK matches). **Opposite** for surprises: US negative at the long end, UK positive at the short end — and at 25y, beyond the US range, the UK flips regime again: liquidity dies, surprises return significant.
- **RQ3** US: CiP dominates (0.39). UK: cointegration dominates ($t-4.4$), CiP weakest. Forwards cointegrated in both; the relation *does* predict. Realised inflation predicts in both (negative sign).

The thread

The **US** is an **inflation-driven** market: surprises move λ (RQ2) and the inflation-aware factor predicts returns (RQ3). The **UK** is a **structure-driven** market: surprises do not bite at the long end, the basis there vanishes, and predictability comes from the forward structure. Both facts point at the *same* long-end clientele.

Declared limits

- The **LDI / clientele** reading of the UK long end and the **protection-demand** reading of the surprise channel are *interpretations*, not tests: no inflation-*fear* measure distinct from the realised surprise, no flow or holdings data in the dataset yet.
- The **deflation-floor channel** is proxied, not measured: its value depends on cumulated inflation since issue, so the correct test is *cross-sectional* by ISIN.
- Indexation lags (3m US; 3m/8m UK) and tax treatment are not modelled.
- UK $E[\pi]$: household long anchor with estimated bias; professional check (MaPS) only from 2023.

Natural next steps

- ISIN-level cross-section for the floor, and holdings data for the clientele story.
- **Extend to euro-area linkers** (Italy, Germany, France) — three sovereign issuers indexed to the *same* euro-area inflation (HICP_{xT}, CPTFEMU), a natural setting to separate credit from inflation-linked pricing.